Hints: Linear algebra – 7. Euclidean spaces, orthogonality and the SVD

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. First show that a real symmetric A has a real eigenvalue. Take a complex eigenvector x and compare $\bar{x}^T A x$ with its conjugate. Then show that $x^\perp$ is invariant ($\langle Ay, x\rangle = \langle y, Ax\rangle$) and induct on the dimension.

Hint 2. In an orthonormal eigenbasis, $\langle Ax, x\rangle = \sum \lambda_i x_i^2 \leqslant \lambda_{\max} \sum x_i^2$. Equality holds at a top eigenvector.

Hint 3. A normal matrix is $A = UDU^*$ with U unitary. If D is real, then $A^* = A$. A real Hermitian matrix is symmetric.

Problem 2

(a)

Hint 1. Put $P = AA^*$ and $Q = A^*A$, both positive semidefinite. They have the same eigenvalues (AB vs BA, same size). Also $Q = P^2$ has the squares of the eigenvalues of P.

Hint 2. So the multiset of eigenvalues $\mu_i \geqslant 0$ of P is invariant under $\mu \mapsto \mu^2$. Look at the largest eigenvalue and the smallest positive one to get $\mu_i \in \{0, 1\}$. Then $P^2 = P$, i.e. $A^*A = P^2 = P = AA^*$.

(b)

Hint 1. A is normal, so it is unitarily diagonalisable, and AA^* has eigenvalues $|\lambda|^2$. These lie in $\{0, 1\}$.

Problem 3

Hint 1. $u^T A u = 0$ for all u means the quadratic form of A vanishes. So the symmetric part $\frac{1}{2}(A + A^T)$ is 0.

Hint 2. Write a 3×3 skew-symmetric matrix as $\begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}$ and compare with $v \times u$ for $v = (a, b, c)$.

Problem 4

Hint 1. Let R be the first k rows. Complete them to some orthogonal matrix $\begin{pmatrix} R \\ R' \end{pmatrix}$. Any $U \in O(n-k)$ gives another completion $\begin{pmatrix} R \\ UR' \end{pmatrix}$. Choose U so that the first l columns come out right.

Hint 2. Write X for the common $k \times l$ corner, Z for the prescribed lower part of the first l columns, and R'_l for the first l columns of R'. Orthonormality of the columns gives $R_l'^T R'_l = E - X^T X = Z^T Z$.

Hint 3. Two matrices with the same Gram matrix differ by an isometry between their column spaces. Extend it to $U \in O(n-k)$ with $UR'_l = Z$.

Problem 5

Hint 1. Multiply by R^T: it suffices to rule out $P' + Q' = E$ with P', Q' orthogonal. Then $E - P'$ is orthogonal; expand $(E-P')^T(E-P') = E$.

Hint 2. This gives $P' + P'^T = E$, so $x^T P' x = \frac{1}{2}|x|^2$ for all real x. But a real 3×3 orthogonal matrix has a real eigenvector with eigenvalue ± 1 (odd dimension), and there $x^T P' x = \pm|x|^2$. Contradiction. In dimension 2 it does work: rotation by $60°$.